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pseudo code, flow chart and dry run.

write pseudo code and draw flow chart for given problem and explain the pseudo code with proper dry run.

① check given number is palindrome or not.

pseudo code:

Start.

Read (number).

set originalNumber = number.

set reverse = 0.

while number > 0 Do.

 remainder = number MOD 10.

 reverse = (reverse * 10) + remainder

 number = number DIV 10.

End while.

If originalNumber == reverse Then.

 print "Number is palindrome."

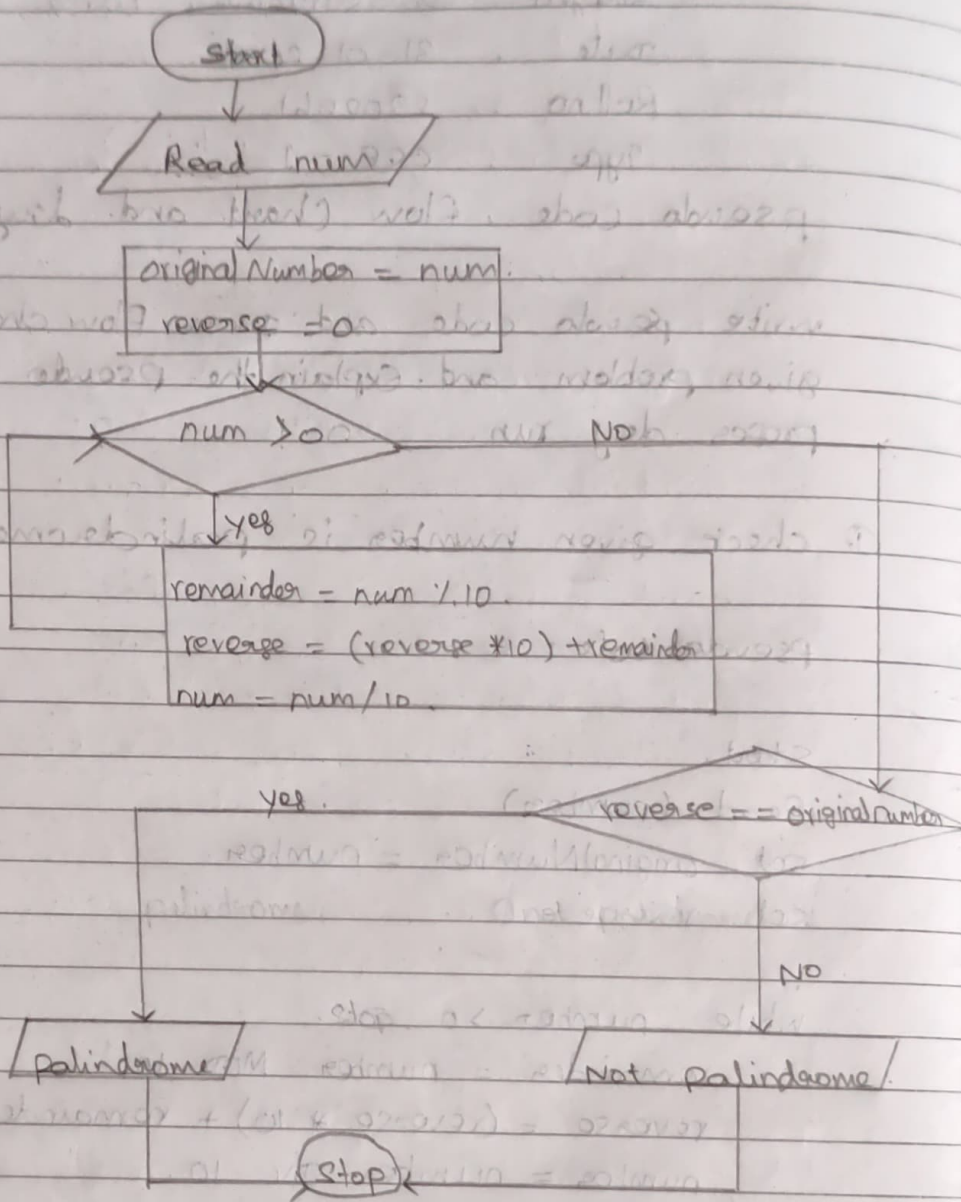
ELSE

 print "Number is not palindrome"

end if

Stop.

Flowchart :



Dry run :

Steps	Number	remainder	Reverse
initial	121	0	0
1	12	1	1
2	1	2	12
3	0	1	121

So that original Number and Reverse are equal and 121 is palindrome.

Q check given number is perfect or not.

Pesudo code :

Start

Read number

set sum = 0

For i = 1 To number / 2 DO

IF number MOD i == 0 THEN

sum = sum + i

end if

end For

IF sum == number THEN

print "number is perfect"

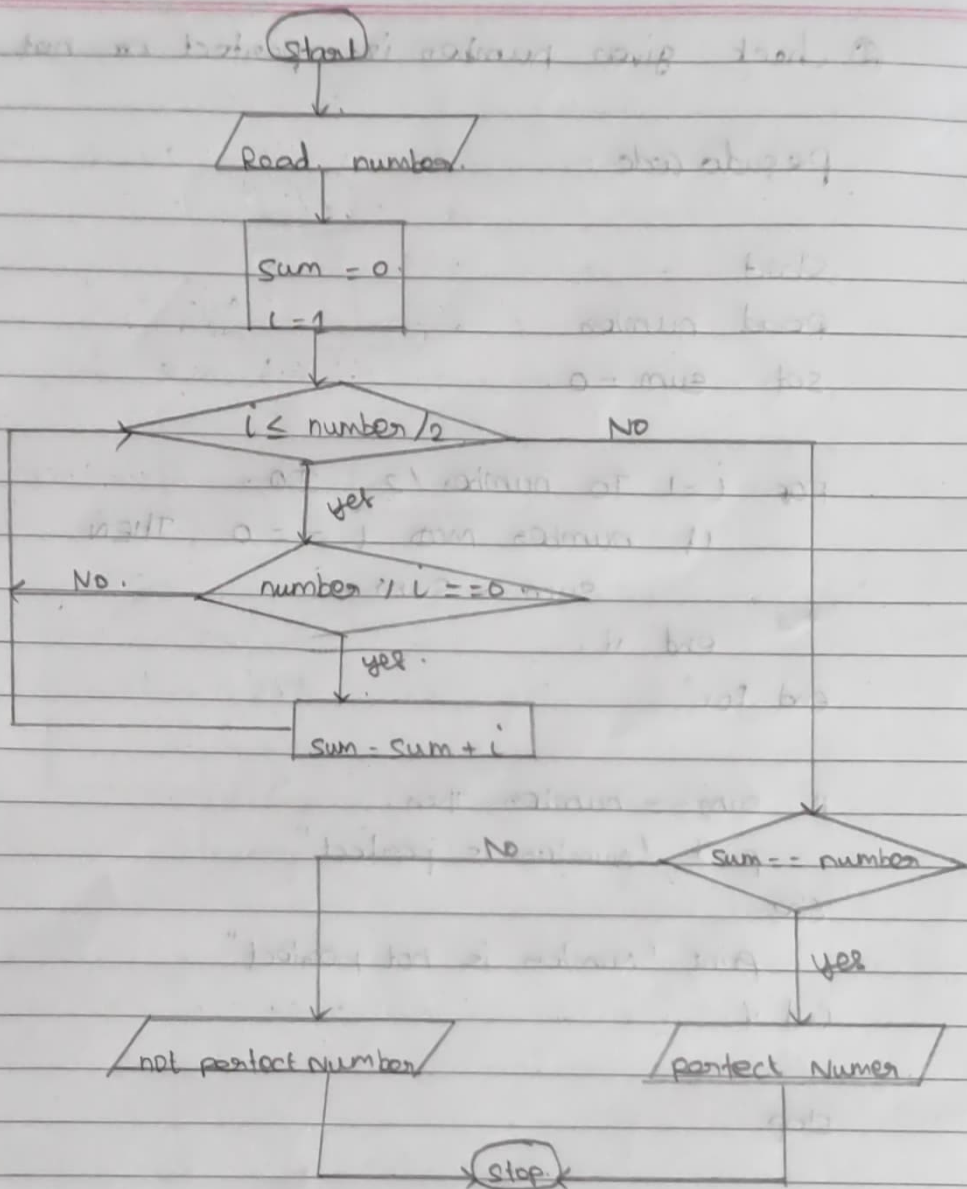
Else

print "number is not perfect"

End if

Stop

Flow chart



Dry Run.

Input : 6.

i	number % i	Sum
1	6 % 1 = 0	1
2	6 % 2 = 0	3
3	6 % 3 = 0	6 //
4 X		

Final sum = 6.

6 is a perfect number.

- ③ write a ~~per~~ pseudo code to insert element in between array.

Pseudo code:

Start.

Read n .

Declare Array A of size $n+1$

For $i=0$ to $n-1$ do.

 Read $A[i]$.

end For.

Read element.

Read position.

For $i=n$ Down to position do.

$A[i] = A[i-1]$.

end For.

$A[\text{position}] = \text{element}$.

$n = n+1$.

print "Array after insertion"

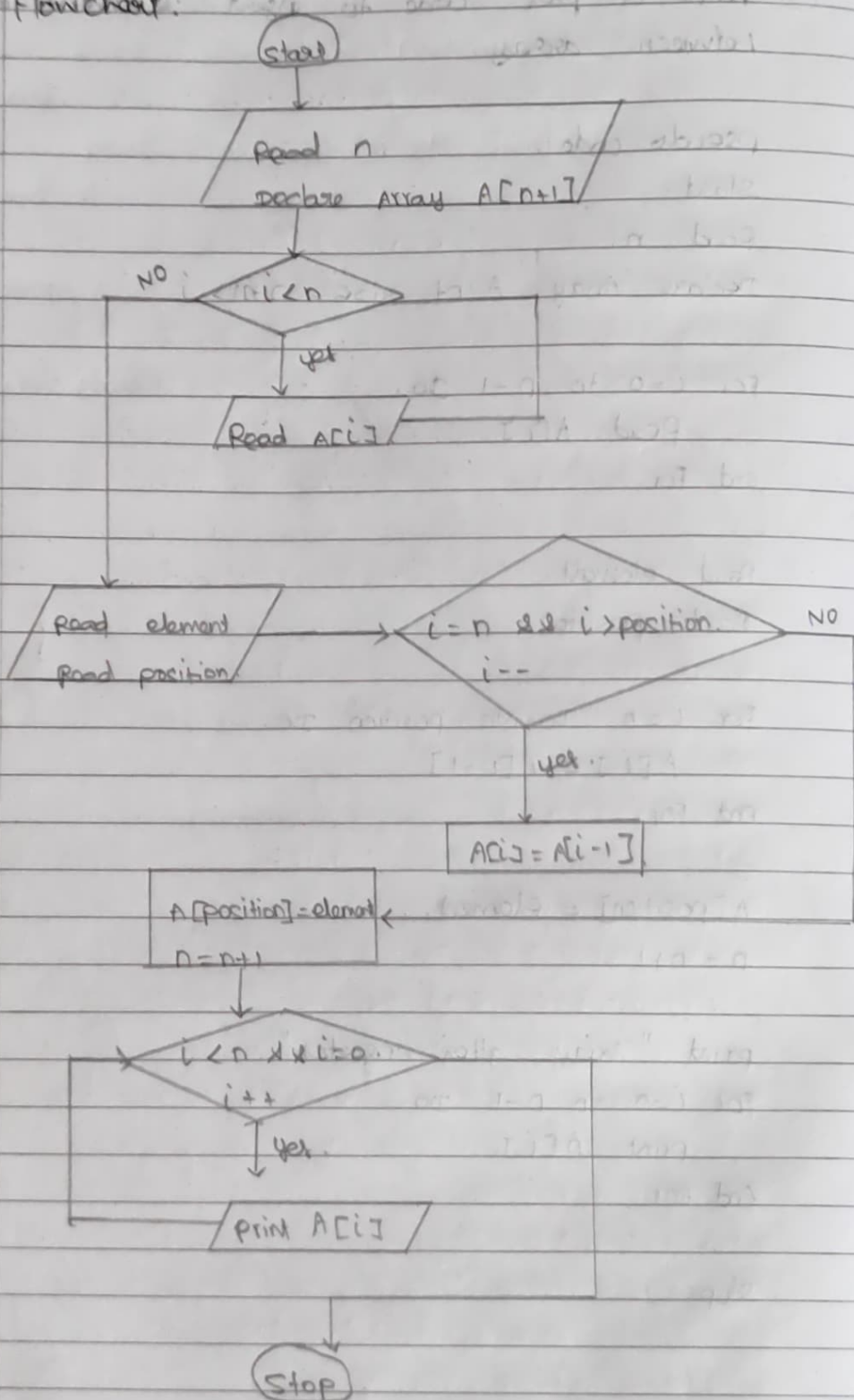
For $i=0$ to $n-1$ do.

 print $A[i]$

end For.

Stop.

Flowchart:



Dry Run:

initial array: $[5, 2, 3]$

element to insert: 8

position: 3

Step	Array
Before insert	$[5, 2, 3]$
Shift 3	$[5, 2, 3, 3]$
insert 8	$[5, 2, 8, 3]$

① Check whether given number is armstrong or not.

pseudo code:

Start

Read number.

Set originalNumber = number.

Set sum = 0.

Set count = 0.

Set temp = number

while temp > 0 Do.

 count = count + 1

 temp = temp Div 10

end while.

Set temp = number.

while temp > 0 Do.

 digit = temp Mod 10.

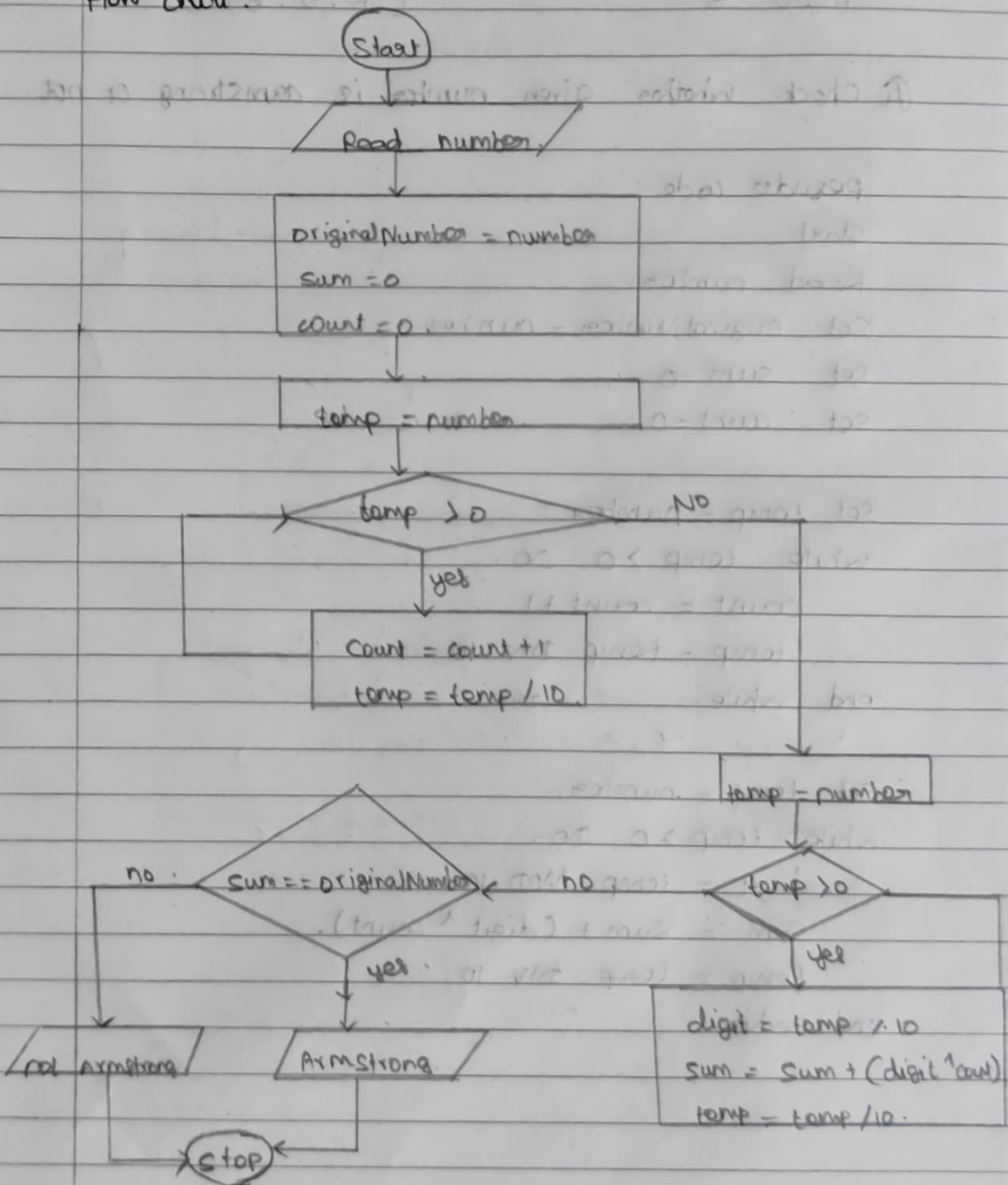
 sum = sum + (digit ^ count).

 temp = temp Div 10.

end while.

if $sum == OriginalNumber$, then
 Print "Number is Armstrong"
 else
 print. "Number is not Armstrong"
 end if.
 stop.

Flow chart :



my Run

input : 153.

number of digit : 3.

Digit	Calculation	Sum
3	$3^3 = 27$	27.
5	$5^3 = 125$	152.
1	$1^3 = 1$	153 //

153 is a Armstrong number

- ⑤. Accept array elements and find Maximum and minimum elements from the array.

Pseudo code :

Start

Read n.

Declare Array A of size n.

For $i = 0$ To $n-1$ Do.

Read $A[i]$.

end for.

set $MIN = 0$

set $MAX = 0$.

For $i = 0$ To $n-1$ Do.

If $A[i] > MAX$ then.

$MAX = A[i]$.

End If.

If $A[i] < MIN$ then.

$MIN = A[i]$

end If.

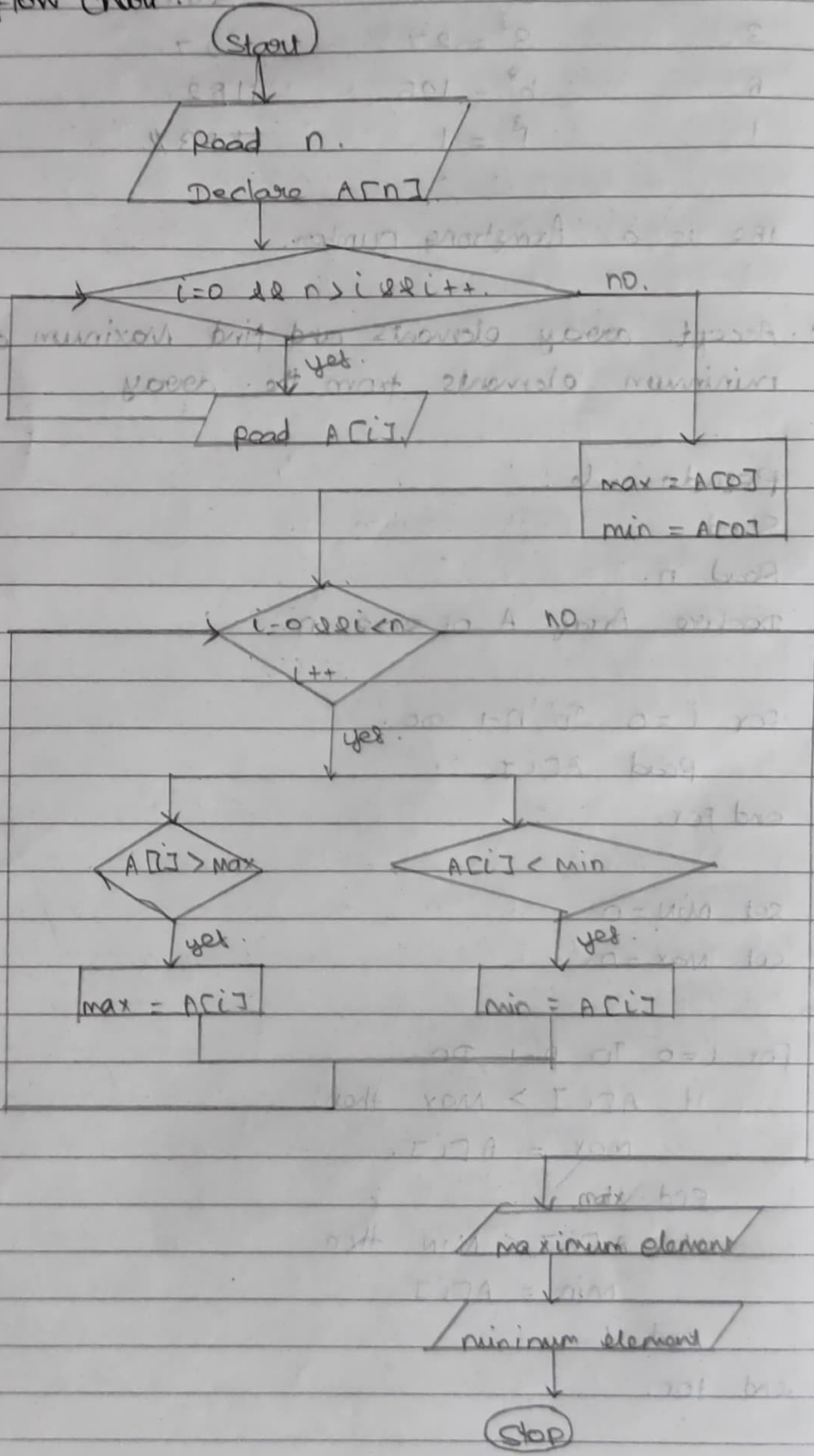
end for.

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print "Maximum element =", max.
print "Minimum element =", min.
stop.

```

Flow chart :



Dry Run:

Array elements: [12, 45, 7, 32, 19]

element	max	min
Start.	12	12
45	45	12
7	45	7
32	45	7
19	45//	7//

The given array.

Minimum element is 7

Maximum element is 45.